

IBM BOB AI HACKATHON • AI
TRACK

Supply Chain Disruption Assistant & Cold Chain Monitor

Real-Time Disruption Resolution & Precision Temperature Governance

Powered by IBM watsonx.ai • Granite-3-8b-instruct

Team Catalyst

Hitarth Hindocha (Lead) | Harsh Vora | Atibali Saiyed | Bhakti Moteriya

 supplychain-catalyst-production.up.railway.app

 Spring Boot 3.3 + Java 17

 IBM watsonx.ai Granite

 Custom Dijkstra Engine

 H2 In-Memory Database

 Railway Cloud Deployment

Supply Chain Vulnerability & Unseen Cold Chain Excursions

The operational & financial cost of delayed visibility in modern logistics

01

Invisible Cascade Effect

Severe weather, port strikes & geopolitical crises cascade across hundreds of active shipments simultaneously. Logistics managers lack real-time automated mapping, forcing manual tracking and delayed reactions after disruptions occur.

02

Critical Cold Chain Risk

Vaccines, biopharmaceuticals & perishables require strict uninterrupted thermal control. A single undetected temperature excursion can instantly spoil cargo worth \$500,000+ in transit — with no warning.

03

Delayed Post-Facto Discovery

Temperature breaches are only discovered upon physical arrival at destination. By the time cargo is unsealed, it is too late to intervene, reroute, or rescue perishable assets from spoilage.

Team Catalyst End-to-End Operational Intelligence

Proactive impact matching, graph-based rerouting, and AI Ops synthesis

1 Disruption Impact Detection

Matches active weather, strike & geopolitical disruptions against live shipment routes to identify affected cargo immediately.

2 Graph Dijkstra Rerouting

Runs Dijkstra's shortest-path algorithm across Indian logistics hub network to calculate optimal alternate routes and match idle fleet.

3 Cold Chain Monitoring

Classifies real-time temperature sensor excursions by regulatory severity (Minor vs Major breach) for instant intervention.

4 IBM Granite AI Ops Brief

IBM watsonx.ai Granite model transforms complex live multi-domain data into clear, plain-English executive briefings for managers.

System Components & Data Flow

Spring Boot 3.3 • Custom Dijkstra Engine • IBM watsonx.ai REST API



RouteOptimizationService.java

Dijkstra graph engine
Custom priority queue
Adjacency list model

DisruptionService.java

Segment/node matching
Affected shipment detection
Blocked edge resolution

ColdChainService.java

Breach classification
Latest reading lookup
Owner notification

WatsonxService.java

IAM token exchange
WebClient REST calls
3-tier fallback logic

Dijkstra Route Optimization in Action

NH48 Weather Disruption → 12 Shipments Rerouted → Optimal Bypass Calculated

01 Disruption Logged

NH48 blocked by severe monsoon flooding — HIGH severity

02 Impact Detected

System identifies 12 shipments routed through NH48

03 Dijkstra Runs

Graph algorithm computes optimal bypass in real-time

04 Dispatcher Acts

Reviews +90 km / +1.8h delta, clicks Approve Reroute

✗ **DISRUPTED ROUTE (BLOCKED)** — 1,420 km

MUMBAI_PORT → SURAT_HUB → JAIPUR_CORRIDOR → DELHI_HUB

Blocked by: NH48 (Severe Monsoon Flooding)

✓ **DIJKSTRA BYPASS ROUTE** — 1,510 km (+90 km)

MUMBAI_PORT → NASHIK_HUB → GWALIOR → DELHI_HUB

Est. Delay: +1.8 hours

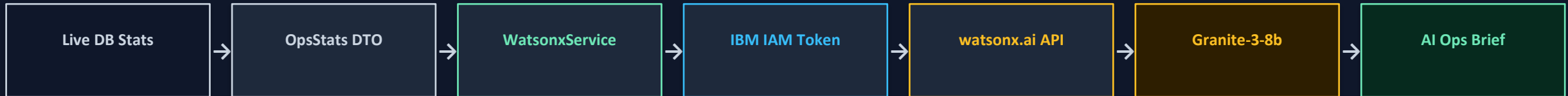
Bypass Highway: NH160_BYPASS

Recommended Carrier: Safexpress (Score: 93.1/100)

Action: Approve Reroute ← 1-click per shipment

IBM watsonx.ai & Granite 3 Model Integration

Converting raw operational metrics into plain-English tactical insights



🔑 Credentials & API Setup

1. IBM Cloud IAM — API key generation
2. Watson Machine Learning Lite plan
3. watsonx.ai Project ID configuration
4. Spring Boot env: WATSONX_API_KEY + WATSONX_PROJECT_ID
5. Model: ibm/granite-3-8b-instruct

🛡️ Architectural Resilience

- Zero-Crash Guarantee: falls back to in-memory rule engine if credentials missing
- 3-Tier Priority: watsonx.ai → Ollama → System Fallback
- Graceful UI Banner: 'AI unavailable — fallback brief'
- IAM Token Cache: auto-refreshes before expiry

📄 AI Ops Brief Output

Granite synthesises 6 live metrics:

- Active shipments & disruption count
- Affected shipment tally
- Idle fleet availability
- Cold-chain breach severity

→ Returns prioritised 5-section executive brief

Measured Outcomes & Platform Performance

Real numbers from the live demo environment running on Railway

OPERATIONAL METRICS

- 18 shipments tracked across 6 freight corridors
- 4 active disruption events monitored
- 12 shipments rerouted on NH48 event alone
- 1-click reroute approval per shipment
- Owner notification on delay / breach

DIJKSTRA RESULTS (NH48)

Original route: 1,420 km (blocked)
Bypass route: 1,510 km (+90 km)
Est. delay added: +1.8 hours
Bypass path: NH160_BYPASS
Carrier match: Safexpress 93.1/100

COLD CHAIN MONITORING

- 7 cold-chain loads under live telemetry
- 2 Major Breach alerts auto-detected
- 1 Minor Breach flagged automatically
- 0 manual interventions required
- Notify Owner flow per breach

IBM WATSONX.AI

- Live OpsStats → Granite-3-8b-instruct
- Plain-English brief in < 2 seconds
- 3-tier fallback: zero crash guarantee
- Load-bearing AI — not decorative
- watsonx REST API fully integrated

Team Catalyst — IBM Bob AI Hackathon • AI Track

Supply Chain Disruption Assistant & Cold Chain Monitor

Hitarth Hindocha

Team Lead & Backend Engineer

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- Backend architecture & Spring Boot core
- Custom Dijkstra engine (RouteOptimizationService)
- IBM watsonx.ai integration & IAM auth
- DisruptionService logic & bug fixes

Harsh Vora

Frontend & UI Engineer

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- Thymeleaf HTML templates & dark-mode CSS
- Disruption impact page design
- Shipment tracking portal
- Dashboard & cold chain UI

Atibali Saiyed

Data & Services Engineer

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- ColdChainService & temperature breach logic
- DataSeeder & demo dataset design
- JPA entities & repository layer
- Fleet asset module

Bhakti Moteriya

Docs, Deployment & Design

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- docs/ documentation suite
- submission.yaml & hackathon checklist
- Presentation design & slide deck
- Railway cloud deployment